SUPERSEDING
See Section 6

MILITARY SPECIFICATION

CONTROL UNIT: LANCE AK

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of subassembly, entitled Control Unit, which is used in the Lance warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-48305	- Squib Switches.
MIL-G-48529	- Rate Gyroswitch
MIL-P-48534	- Printed Wiring Board.
MIL-D-48522	- Delay Unit.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

NUOR

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DRAWINGS

PICATINNY ARSENAL
PRODUCT AND PACKAGING DRAWINGS

9236970	- Plate Mounting (Casting).
9236980	- Mounting Plate Assembly.
9236990	- Cover.
9237001	- Control Unit.
ET-9237001	- Equipment Tabulation.
9237017	- Plate, Connector.
9237019	- Plate, Connector Assembly.
9237021	- Terminal Board and Harness Assembly.
9237022	- Printed Wiring Board Assembly.
9237028	- Terminal Board, Component in Jumper Installation.
9237302	- Energy Transfer Module (ETM).

PUBLICATIONS

PICATINNY ARSENAL

QAI-PA-1 - Quality Assurance Instructions for Reporting Data on Nuclear Weapons.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The Control Unit shall comply with all requirements specified on the drawing (dwg) 9237001, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-N-A. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

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3.5 Coded serialization requirements (see 6.1.2). The contractor shall supply feedback data on punch cards for any item for which coding is required in section 3. The feedback data shall be in accordance with QAI-PA-1.

3.6 Components. For components to be used in the Control Unit, objective evidence (see 6.2) indicating that the following components conform to their respective specifications and/or drawings shall be made available to the Government representative:

a. M46 Rate Gyroswitch - MIL-G-48529(PA).
b. Squib Switches - MIL-S-48305(PA).
c. Printed Wiring Board - MIL-P-48534(PA).
d. Delay-Unit - MIL-D-48522(PA).
e. Plate Mounting (Casting) - Drawing 9236970.
f. Energy Transfer Module (ETM) - MIL-M-63174

3.7 Circuit resistance - PWB #1 Assembly dwg 9237022 (Before Conformal Coating (BCC) and After Conformal Coating (ACC): The resistance of each circuit shall not exceed the maximum value listed in Table I or as shown otherwise when tested as specified in 4.4.1.

Table I

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J19-1A	J19-4B	0.3	060001
J19-1B	J19-4A	0.3	060002
J19-1C	J19-1F	0.3	060003
J19-1D	J19-1E	0.3	060004
J19-1G	J19-4H	0.3	060005
J19-1G	J19-4J	0.3	060006
J19-1K	J19-4G	1.5-2.5	060007
J19-1K	J19-4F	24-32	060008
J19-2A	J19-4K	0.3	060009
J19-2A	J19-2W	12-16	060010
J19-2B	J19-3B	0.3	060011
J19-2C	J19-2D	1.5-2.5	060012
J19-2D	J19-2E	10-14	060013
J19-2E	J19-4F	0.3	060014
J19-2F	J19-5A	0.3	060015
J19-2H	J19-2J	0.3	060016
J19-2G	177	0.3 (BCC only)	060017
J19-2K	181	0.3 (BCC only)	060018
J19-5C	J19-5D	1.5-2.5	060019
J19-3A	J19-4S	0.3	060020
J19-5F	159	0.3 (BCC)	060021
J19-3C	J19-3E	0.3	060022

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Table I (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J19-3C	J19-4F	12-16	060023
J19-3D	J19-3F	0.3	060024
J19-1S	J19-2M	1.5-2.5	060025
J19-1L	126	0.3 (BCC only)	060026
J19-1M	122	0.3 (BCC only)	060027
J19-1N	121	0.3 (BCC only)	060028
J19-1P	120	0.3 (BCC only)	060029
J19-1R	119	0.3 (BCC only)	060030
J19-1S	J19-1T	1.5-2.5	060031
J19-1S	J19-1V	12-16	060032
J19-1S	J19-2L	12-16	060033
J19-1S	J19-2N	2.5-4.0	060034
J19-1U	124	0.3 (BCC only)	060035
J19-1V	J19-3G	0.3	060036
J19-1V	J19-3H	0.3	060037
J19-5E	149	0.3 (BCC)	060038
J19-1W	130	0.3 (BCC only)	060039
J19-2M	J19-2N	1.5-2.5	060040
J19-2L	J19-1V	0.3	060041
J19-2P	132	0.3 (BCC only)	060042
J19-2R	133	0.3 (BCC only)	060043
J19-2S	134	0.3 (BCC only)	060044
J19-2T	135	0.3 (BCC only)	060045
J19-2U	131	0.3 (BCC only)	060046
J19-2V	J19-3P	0.3	060047
J19-3M	J19-3R	0.3	060048
J19-3M	J19-3S	12-16	060049
J19-5K	J19-5J	0.3	060050
J19-3L	J19-3N	0.3	060051
J19-3U	J19-4P	0.3	060052
J19-3V	J19-4M	0.3	060053
J19-3W	J19-4L	0.3	060054
J19-3V	J19-4N	0.75-2	060055
J19-4C	J19-4E	24-32	060056
J19-4C	J19-4F	0.3	060057
J19-4D	J19-4E	1.5-2.5	060058
J19-4F	J19-4N	12-16	060059
J19-4T	J19-4U	1.5-2.5	060060
J19-4V	J19-4U	0.3	060061
J19-4V	J19-4W	1.5-2.5	060062
J19-5B	J19-4F	0.3	060063
J19-5B	J19-5C	10-14	060064
J19-1V	J19-3B	4-8	060065
J19-5L	191	0.3 (BCC)	060066
J19-5M	192	0.3 (BCC)	060067

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3.8 Insulation resistance - printed wiring board assembly dwg 9237022. The insulation resistance between each circuit and all other circuits, as indicated in Table II, shall not be less than 100 megohms when tested as specified in 4.4.2.

Table II

From	To	From	To
J19-1A	All others listed & shell	J19-2K	All others listed & shell
J19-1B	"	J19-2P	"
J19-1C	"	J19-2R	"
J19-1D	"	J19-2S	"
J19-1G	"	J19-2T	"
J19-1L	"	J19-2U	"
J19-1M	"	J19-2V	"
J19-1N	"	J19-3D	"
J19-1P	"	J19-3L	"
J19-1R	"	J19-5J	"
J19-1U	"	J19-3U	"
J19-1W	"	J19-3W	"
J19-2F	"	J19-3A	"
J19-2G	"	J19-5E	"
J19-2H	"	J19-5F	"
		J19-5L	"
		J19-5M	"

The following list of unused pins may be connected and tested as one terminal: 1H, 1J, 3J, 3K, 5G, 5H, 3T, 4R, 5R, 5S, 5T, 5U, 5V, 5W, 5P, 5N.

3.9 Circuit resistance - term, bd harness assembly dwg 9237021 (BCC and ACC). The resistance of each circuit shall not exceed the maximum value listed in Table III, or as shown otherwise when tested as specified in 3.8.1, 3.8.2 and 4.4.1.1.

3.9.1 Relay coil test voltage polarity. Certain test codes listed in Table III require that a relay be energized with 30 + 2v to perform the test. When a relay "ON" requirement is listed for a test code, the relay shall be energized by a floating deck power supply connected as follows:

Relay Desig	Polarity	
	+	-
K1A	P19-5E	P19-2W
K1B	P19-5F	P19-2W
K2A	1J2-a	1J2-D
K2B	1J2-$\overline{S}$	1J2-U

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3.9.2 Test terminal polarity. Certain test codes require a voltage of proper polarity to be used to obtain the test result indicated in Table III. Test voltage polarity shall be used, as indicated by symbols appearing in Table III, as (+) or (-) following the terminal to which they apply.

Table III

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-1A	P19-4J	0.3	080001	
P19-1A	P8-H	0.3	080002	
P19-1A	P8-K	0.3	080003	
P19-1A	P8-J	0.3	080004	
P19-1A	P8-L	0.3	080005	
P19-1B	P19-4H	0.3	080006	
P19-1B	P9-H	0.3	080007	
P19-1B	P9-J	0.3	080008	
P19-1B	P9-K	0.3	080009	
P19-1B	P9-L	0.3	080010	
P19-1C	P7-J	0.3	080011	
P19-1C	P7-H	0.3	080012	
P19-1D	P6-J	0.3	080013	
P19-1D	P6-H	0.3	080014	
P19-1E	J3-c	0.3	080015	
P19-1E	CR203-A	0.3 (BCC)	080016	
P19-1F	J3-w	0.3	080017	
P19-1F	CR202-A	0.3 (BCC)	080018	
P19-1G	J3-L	21-33	080019	
P19-1K	P5-29	0.3	080020	
P19-1L	J3-AA	0.3	080021	
P19-1M	1J1-D	0.3	080022	
P19-1M	1J1-E	0.3	080023	
P19-1M	CR-410-C	0.3 (BCC)	080024	
P19-2B	1J12-G	0.3	080025	
P19-2B	1J12-H	0.3	080026	
P19-2B	1J12-J	0.3	080027	
P19-2B	1J12-X	0.3	080028	
P19-2B	P5-31	0.3	080029	
P19-1N	1J1-M	0.3	080030	
P19-1N	1J13-62	0.3	080031	
P19-1N	P19-2S	0.3	080032	
P19-1N	1J1-L	0.3	080033	
P19-1P	1J1-K	0.3	080034	
P19-1P	1J1-b	0.3	080035	
P19-1P	P19-2R	0.3	080036	
P19-1P	1J13-63	0.3	080037	
P19-1R	P4-Z	0.3	080038	
P19-1S	P5-23	0.3	080039	
P19-1T	P5-51	0.3	080040	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-1U	J3-v	0.3	080041	
P19-1V	P4-$\overline{A}$A	0.3	080042	
P19-1V	P4-BB	0.3	080043	
P19-1V	P19-2L	0.3	080044	
P19-1W	J3-d	0.3	080045	
P19-2A	J15-$\overline{M}$	0.3	080046	
P19-2C	CR205-C	0.3 (BCC)	080047	
P19-2C	P5-7	0.3	080048	
P19-2C	P8-P	0.3	080049	
P19-2C	P9-U	0.3	080050	
P19-2C	1J13-9	0.3	080051	
P19-2D	P9-M	0.3	080052	
P19-2D	P8-N	0.3	080053	
P19-2D	P5-5	0.3	080054	
P19-2D	P19-2E	5-7	080055	
P19-2G	1J1-C	0.3	080057	
P19-2G	1J1-S	0.3	080058	
P19-2G	1J1-V	0.3	080059	
P19-2G	1J1-d	0.3	080060	
P19-2G	1J2-$\overline{L}$	0.3	080061	
P19-2G	1J2-V	0.3	080062	
P19-2G	P4-V	0.3	080063	
P19-2G	P4-c	0.3	080064	
P19-2G	P19-$\overline{2K}$	0.3	080065	
P19-2H	P8-E	0.3	080066	
P19-2H	P8-B	0.3	080067	
P19-2H	P8-C	0.3	080068	
P19-2H	P8-D	0.3	080069	
P19-2H	P9-B	0.3	080070	
P19-2H	P9-C	0.3	080071	
P19-2H	P9-D	0.3	080072	
P19-2H	P9-E	0.3	080073	
P19-2H	1J13-25	0.3	080074	
P19-2H	P19-2J	0.3	080075	
P19-2H	J3-E	0.3	080076	
P19-2M	P5-45	0.3	080077	
P19-2M	1J13-15	0.3	080078	
P19-2N	P5-22	0.3	080079	
P19-2P	P4-b	0.3	080080	
P19-2U	J3-$\overline{P}$	0.3	080081	
P19-2V	1J2-M	0.3	080082	
P19-2T	1J1-W	0.3	080083	
P19-2T	1J1-X	0.3	080084	
P19-2T	CR412-C	0.3 (BCC)	080085	
P19-2W	P6-B	0.3	080086	
P19-2W	P6-C	0.3	080087	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-2W	P6-E	0.3	080088	
P19-2W	P6-G	0.3	080089	
P19-2W	P7-B	0.3	080090	
P19-2W	P7-C	0.3	080091	
P19-2W	P7-E	0.3	080094	
P19-2W	P5-9	0.3	080095	
P19-2W	P5-16	0.3	080096	
P19-2W	P5-47	0.3	080097	
P19-2W	P5-48	0.3	080098	
P19-2W	P4-M	0.3	080099	
P19-2W	P4-N	0.3	080100	
P19-2W	1J13-24	0.3	080101	
P19-2W	P19-3S	0.3	080102	
P19-2W	TD201-1	0.3 (BCC)	080105	
P19-2W	E-202	0.3 (BCC)	080106	
P19-2W	P7-D	0.3	080107	
P19-2W	P7-F	0.3	080108	
P19-2W	P6-F	0.3	080109	
P19-2W	P6-D	0.3	080110	
P19-2W	P7-A	14-16	080111	
P19-2W	P7-L	14-16	080112	
P7-T	P7-K	0.3	080113	
P6-T	P6-K	0.3	080115	
P19-2W	P6-A	14-16	080117	
P19-2W	P6-L	14-16	080118	
P19-3A	1J13-44	0.3	080119	
P19-3B	1J12-F	0.3	080120	
P19-3B	1J12-Y	0.3	080121	
P19-3B	1J12-Z	0.3	080122	
P19-3B	1J12-f	0.3	080123	
P19-3B	P5-30	0.3	080124	
P19-5K	1J2-F	0.3	080125	
P19-5J	1J2-W	0.3	080126	
P19-3W	1J16-S	0.3	080127	
P19-3W	1J17-S	0.3	080128	
P19-3C	1J2-E	0.3	080129	
P19-3C	1J13-6	0.3	080130	
P19-3D	P10-V	0.3	080131	
P19-3E	1J2-Y	0.3	080132	
P19-3F	P10-W	0.3	080133	
P19-3F	1J13-5	0.3	080134	
P19-3G	J3-HH	0.3	080135	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-2W	P10-D	0.3	080137	
P19-2W	P10-N	0.3	080138	
P19-2W	P7-G	0.3	080139	
P19-3H	J3-u	0.3	080141	
P19-3L	1J2-A	0.3	080142	
P19-3M	J3-M	0.3	080143	
P19-3M	P19-3R	0.3	080144	
P19-3M	1J13-3	0.3	080145	
P19-3M	CR202-C	0.3 (BCC)	080146	
P19-3N	1J2-J	0.3	080147	
P19-3P	1J2-G	0.3	080148	
P19-3U	P5-25	0.3	080149	
P19-3V	P19-4M	0.3	080150	
P19-3V	P5-50	0.3	080151	
P19-3V	1J13-27	0.3	080152	
P19-3W	P19-4L	0.3	080153	
P19-4A	CR208-A	0.3 (BCC)	080154	
P19-4A	CR207-A	0.3 (BCC)	080155	
P19-4B	CR209-A	0.3 (BCC)	080156	
P19-4B	CR210-A	0.3 (BCC)	080157	
P19-4C	P4-X	0.3	080158	
P19-4C	P19-4F	0.3	080160	
P19-4C	P10-Y	0.3	080163	
P19-4C	TD201-8	0.3 (BCC)	080164	
P19-4C	P4-Y	0.3	080165	
P19-4C	P10-Z	0.3	080166	
P19-4C	P5-12	8-12	080167	
P19-4C	P5-11	8-12	080168	
P19-4D	P19-4G	0.3	080169	
P19-4D	P5-20	0.3	080170	
P19-4D	P5-21	0.3	080171	
P19-4D	1J13-8	0.3	080172	
P19-4D	CR205-A	0.3 (BCC)	080173	
P19-4E	P5-28	0.3	080174	
P19-4N	P5-49	0.3	080175	
P19-4P	P5-4	0.3	080176	
P19-4K	J15-L	0.3	080177	
P19-4S	J3-J	0.3	080178	
P19-4S	1J13-49	0.3	080179	
P19-4T	P5-52	0.3	080180	
P19-4U	P5-18	0.3	080181	
P19-4V	P5-19	0.3	080182	
P19-4V	1J13-14	0.3	080183	
P19-4W	P5-46	0.3	080184	
P19-5B	P19-5C	5.4-6.9	080186	
P19-5B	P5-8	5.4-6.9	080187	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P19-5B	P9-N	5.4-6.9	080188	
P19-5B	P8-M	5.4-6.9	080189	
P19-5D	P5-6	0.3	080190	
P19-5D	P9-P	0.3	080191	
P19-5D	P8-U	0.3	080192	
P19-5D	1J13-10	0.3	080193	
P19-5E	J15-R	0.3	080194	
P19-5E	1J13-35	0.3	080195	
P19-5F	J15-S	0.3	080198	
P19-5F	1J13-39	0.3	080199	
P4-A	1J2-H	0.3	080202	
P4-A	1J13-30	0.3	080203	
P4-B	1J13-29	0.3	080204	
P4-C	P4-D	0.3	080205	
P4-C	J3-R	0.3	080206	
P4-S	P5-26	0.3	080207	
P4-G	P5-27	0.3	080208	
P4-G	P5-33	0.3	080209	
P4-E	1J12-R	0.3	080210	
P4-F	1J12-e	0.3	080211	
P4-E	1J13-22	0.3	080212	
P4-G	1J13-57	0.3	080213	
P4-G	P4-H	0.3	080214	
P4-J	P4-K	0.3	080215	
P4-S	1J13-60	0.3	080216	
P4-T	P4-U	0.3	080217	
P4-W	1J13-28	0.3	080218	
P4-a	P5-24	0.3	080219	
P4-a	1J13-31	0.3	080220	
P4-CC	1J13-18	0.3	080221	
P4-CC	1J12-c	0.3	080222	
P4-CC	P4-DD	0.3	080223	
P4-CC	1J12-N	0.3	080224	
P5-35	P5-43	0.3	080225	
P5-35	P5-36	0.3	080226	
P5-39	P6-S	0.3	080227	
P5-39	1J13-33	0.3	080228	
P5-15	P5-14	540-660	080229	
P6-M	P6-N	0.3	080230	
P6-M	P7-M	24-36	080233	
P6-M	P7-N	24-36	080234	
P6-U	P7-U	0.3	080235	
P6-U	1J2-D	0.3	080236	
P6-U	1J2-X	0.3	080237	
P6-U	1J13-50	0.3	080238	
P6-U	K2A-20	0.3 (BCC)	080239	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
P6-R	P7-S	0.3	080240	
P6-R	1J13-34	0.3	080241	
P6-P	P7-P	0.3	080242	
P6-P	1J2-K	0.3	080243	
P6-P	1J2-U	0.3	080244	
P6-P	K2B-20	0.3 (BCC)	080245	
P7-R	P8-R	0.3	080246	
P7-R	1J13-36	0.3	080247	
P8-S	P9-R	0.3	080248	
P8-S	1J13-37	0.3	080249	
P9-S	P10-a	0.3	080250	
P9-S	1J13-38	0.3	080251	
P10-E	J15-H	0.3	080252	
P10-E	1J13-40	0.3	080253	
P15-D	J15-E	0.3	080258	
J15-D	1J13-48	0.3	080259	
J15-G	1J13-44	0.3	080260	
J15-N	J15-P	0.3	080261	
J15-N	1J13-51	0.3	080262	
1J1-A	1J1-U	0.3	080263	
1J1-B	1J12-K	0.3	080264	
1J1-B	1J13-47	0.3	080265	
1J1-F	1J1-Z	0.3	080266	
1J1-F	J3-j	0.3	080267	
1J1-G	1J1-Y	0.3	080268	
1J1-G	J3-MM	0.3	080269	
1J1-H	1J2-B	0.3	080270	
1J1-H	1J13-42	0.3	080271	
1J1-J	J3-X	0.3	080272	
1J1-N	J3-h	0.3	080273	
1J1-P	J3-W	0.3	080274	
1J1-R	J3-S	0.3	080275	
1J1-T	1J1-e	0.3	080276	
1J1-a	J3-b	0.3	080277	
1J1-c	J3-g	0.3	080278	
1J1-f	J3-f	0.3	080279	
1J1-g	J3-m	0.3	080280	
1J1-h	J3-N	0.3	080281	
1J1-h	J3-Z	0.3	080282	
1J1-j	J3-V	0.3	080283	
1J1-j	J3-e	0.3	080284	
1J2-N	1J2-P	0.3	080285	
1J2-N	J3-D	0.3	080286	
1J2-S	1J13-46	0.3	080287	
1J2-S	K2B-19	0.3 (BCC)	080288	
1J2-a	1J13-43	0.3	080289	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
1J2-a	K2A-19	0.3 (BCC)	080290	
1J12-B	J3-x	0.3	080291	
1J12-B	J3-y	0.3	080292	
1J12-B	J3-t	0.3	080293	
1J12-B	J3-r	0.3	080294	
1J12-B	1J12-V	0.3	080295	
1J12-B	1J13-23	0.3	080296	
1J12-D	1J13-16	0.3	080297	
1J12-D	J3-K	0.3	080298	
1J12-L	1J13-55	0.3	080299	
1J12-L	1J13-54	0.3	080302	
1J12-L	1J13-53	0.3	080303	
1J12-L	1J13-52	0.3	080304	
P19-5L	P4-C	0.3	080305	
P5-14	1J13-12	0.3	080310	
P10-C	1J13-1	0.3	080311	
P10-J	1J13-58	0.3	080312	
P10-M	1J13-61	0.3	080313	
P10-R	1J13-32	0.3	080314	
P10-U	1J13-56	0.3	080315	
P10-b	1J13-64	0.3	080316	
P19-2W	P5-35	0.3	080317	
P19-2W	P5-36	0.3	080318	
P19-2W	P5-43	0.3	080319	
P5-26	E-331	0.3 (BCC)	080320	
P5-33	E-305	0.3 (BCC)	080321	
P5-27	E-305	0.3 (BCC)	080322	
P5-47	E-322	0.3 (BCC)	080323	
P5-48	E-314	0.3 (BCC)	080324	
P4-E	1J1-e	0.3	080328	
P4-E	1J1-T	0.3	080329	
P4-F	1J1-A	0.3	080330	
P4-F	1J1-U	0.3	080331	
1J16-C	1J12-f	80-120	080332	
1J17-C	1J12-Y	80-120	080333	
CR-403-C	CR-410-A	0.3 (BCC)	080334	
CR-406-C	CR-412-A	0.3 (BCC)	080335	
CR-403-G	1J16-C	0.3 (BCC)	080336	
CR-406-G	1J17-C	0.3 (BCC)	080337	
J3-BB	ETM 2-IN	0.3 (BCC)	080338	
J3-z	ETM 1-IN	0.3 (BCC)	080339	
ETM 1-OUT	CR-403-A	0.3 (BCC)	080340	
ETM 1-OUT	1J13-59	0.3 (BCC)	080341	
ETM 2-OUT	CR-406-A	0.3 (BCC)	080342	
ETM 2-OUT	1J13-65	0.3 (BCC)	080343	
ETM 1-C	1J12-F	0.3 (BCC)	080344	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
ETM 2-C	1J12-X	0.3 (BCC)	080345	
1J12-h	J3-T	0.3	080346	
1J12-h	J3-GG	0.3	080347	
1J12-h	1J13-67	0.3	080348	
1J12-E	J3-G	0.3	080349	
1J12-E	1J13-17	0.3	080350	
1J12-g	J3-B	0.3	080351	
1J12-g	J3-DD	0.3	080352	
1J12-g	1J13-66	0.3	080353	
1J13-13	J3-p	0.3	080356	
1J13-13	J3-H	0.3	080357	
J3-F	J3-i	0.3	080360	
J3-U	J3-LL	0.3	080361	
J3-Y	J3-JJ	0.3	080362	
J3-a	J3-NN	0.3	080363	
J3-g	J3-KK	0.3	080364	
J3-k	J3-FF	0.3	080365	
J3-CC	P19-1V	0.3	080368	
J3-EE	P19-1V	0.3	080369	
1J2-C	1J2-R	0.3	080370	
1J12-L	1J13-45	0.3	080371	
1J12-L	J3-A	0.3	080372	
P5-15	1J13-11	0.3	080373	
P5-15	CR211-A	0.3 (BCC)	080374	
P5-15	E203	270-330 (BCC)	080375	
P5-15	1J13-12	540-660	080376	
P5-15	CR212-A	540-660 (BCC)	080377	
J15-A	CR211-C	0.3 (BCC)	080378	
J15-A	CR212-C	0.3 (BCC)	080379	
J15-A	1J13-7	0.3	080380	
J15-A	J15-B	0.3	080381	
1J2-c	P5-41	0.3	080382	
1J2-b	P5-40	0.3	080383	
P19-2F	J3-n	0.3	080384	K1A-ON
P19-1G	E435	0.3 (BCC)	080385	
E436	J3-L	0.3 (BCC)	080386	
P19-4A(+)	J3-p (-)	50-74	080387	
P19-4B(+)	J3-H (-)	50-74	080388	
J3-s	P19-5A	0.3	080389	K1B-ON
P19-3B	E410	0.3 (BCC)	080390	
CR403C(+)	1J1-D (-)	50-74 (BCC)	080391	
P19-2B	E412	0.3 (BCC)	080392	

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
CR406C(+)	P19-2T (-)	50-74 (BCC)	080393	
J15-N	J15-P	0.3	080394	
P6-K (+)	P4-T (-)	50-74	080395	
P7-T (+)	P4-U (-)	50-74	080396	
P19-2E	E442	0.3 (BCC)	080397	
P5-6	CR206C	0.3 (BCC)	080398	
P19-4D	CR206A	0.3 (BCC)	080399	
P19-4D(+)	P5-6 (-)	50-74	080400	
P19-4D(+)	P19-2C (-)	50-74	080401	
P5-8	P9-N	0.3	080402	
P19-5B	E434	0.3 (BCC)	080403	
1J2-S (+)	P7-P (-)	180-220	080404	
P7-P (+)	1J2-S (-)	50-80	080405	
P6-U (+)	1J2-a (-)	50-80	080406	
1J2-a(+)	P6-U (-)	180-220	080407	
1J13-55	1J13-54	1 meg (min)	080408	K2B-ON
1J13-54	1J13-53	1 meg (min)	080409	K2A-ON
1J13-53	1J13-52	1 meg (min)	080410	K1B-ON
1J13-52	1J13-45	1 meg (min)	080411	K1A-ON
P19-1F(+)	J3-M (-)	50-74	080412	
J3-c (+)	1J13-3 (-)	50-74	080413	
J3-C	E444	0.3 (BCC)	080414	K2A-ON
J3-C	E444	0.3 (BCC)	080415	K2B-ON
P6-M	E444	12-17 (BCC)	080416	
P7-M	E444	12-17 (BCC)	080417	
P5-15(+)	J15-A (-)	50-74	080418	
E218	P5-14	0.3 (BCC)	080419	
E218 (+)	J15-B (-)	50-74 (BCC)	080420	
P7-L (+)	P4-K (-)	50-74	080422	
P7-L	P7-A	0.3	080423	
P6-A (+)	P4-K (-)	50-74	080424	
P6-A	P6-L	0.3	080425	
P19-2W(+)	P19-5E (-)	50-80	080426	
P19-5E(+)	P19-2W (-)	180-220	080427	
TD201-3	E201	0.3 (BCC)	080428	
TD201-3	E202	40,000 - 55,000 (BCC)	080429	10 ma max
E213	P5-12	0.3 (BCC)	080430	
E215	P5-11	0.3 (BCC)	080431	
P19-5M	E220	0.3 (BCC)	080432	
P10-S	E219	0.3 (BCC)	080433	K2A-ON
P10-T	E221	0.3 (BCC)	080434	K2A-ON

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Table III (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)	Req'd
K1A-11	P5-35	0.3 (BCC)	080435	K1B-ON
K1A-11	P19-2W	0.3 (BCC)	080436	K1A-ON
P7-M	P7-N	0.3	080437	
J15-D	J15-E	0.3	080438	
P10-S	E219	0.3 (BCC)	080439	K2B-ON
P10-T	E221	0.3 (BCC)	080440	K2B-ON
E219	E220	1.3-2.0 (BCC)	080441	
E220	E221	1.3-2.0 (BCC)	080442	
J3-H	CR209C	0.3 (BCC)	080443	
J3-H	CR208C	0.3 (BCC)	080444	
P5-14	CR212A	0.3 (BCC)	080445	
P19-1F	CR201A	0.3 (BCC)	080446	
P19-1E	CR204A	0.3 (BCC)	080447	
P19-3M	CR203C	0.3 (BCC)	080448	
C406 (+)	C405 (-)	0.3 (BCC)	080449	
C407 (+)	C408 (-)	0.3 (BCC)	080450	
P19-5E	K1A-19	0.3 (BCC)	080451	
P19-4C	K1A-19	0.3 (BCC)	080452	K1A-ON
P19-2W	K1B-9	0.3 (BCC)	080453	K1B-ON
E202	E217	0.3 (BCC)	080454	
E203	E217	0.3 (BCC)	080455	
P7-C	E322	0.3 (BCC)	080456	
P5-36	P5-43	0.3	080457	
P19-2W(+)	P19-5F (-)	50-80	080458	K1A-ON
P19-5F(+)	P19-2W (-)	180-220	080459	K1A-ON
E203	E218	270-330 (BCC)	080460	
K1B-19	P19-4F	0.3 (BCC)	080461	K1B-ON
1J13-60	1J13-24	See Note 1	080464	
1J13-57	1J13-24	See Note 1	080465	
P19-5F	K1B-19	0.3	080466	
P19-2W	K1B-9	1 meg (min) (BCC)	080467	K1A-K1B-ON
K1A-11	P19-2W	1 meg (min) (BCC)	080468	K1A-K1B-ON
1J12-L	1J13-52	1 meg (min)	080469	K1B-ON
K1B-9	P5-43	1 meg (min) (BCC)	080470	K1A-K1B-ON
1J12-L	1J13-53	1 meg (min)	080471	K2A-ON
1J12-L	1J13-53	1 meg (min)	080472	K2B-ON

Note 1: Set Simpson 269 Meter to RX100 Scale. Connect positive lead of Simpson Meter to test points listed in FROM column. Connect negative lead of Simpson Meter

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to test point listed in the TO column. The Simpson Meter will slowly rise to a high resistance.
CAUTION: DO NOT REVERSE POLARITY

3.10 Insulation resistance tests.

3.10.1 100 Megohm insulation resistance terminal board and harness assembly drawing 9237021. The insulation resistance between each circuit and all other circuits as indicated in Table IV shall not be less than 100 megohms when tested as specified in 4.4.2.1.

Table IV

From	To
P19-1A	All others listed & shell
P19-1B	"
P19-1C	"
P19-1L	"
P19-1N	"
P19-1P	"
P19-1R	"
P19-1S	"
P19-1U	"
P19-1W	"
P19-2A	"
P19-2D	"
P19-2F	"
P19-2G	"
P19-2H	"
P19-2M	"
P19-2N	"
P19-2P	"
P19-2U	"
P19-2V	"
P19-3A	"
P19-3C	"
P19-3D	"
P19-3E	"
P19-3F	"
P19-3G	"
P19-3H	"
P19-3L	"
P19-3N	"
P19-3T	"
P19-3U	"
P19-3V	"

From	To
P19-1D	All others listed & shell
P19-1G	"
P19-1K	"
P4-E	"
P4-F	"
P4-W	"
P4-a	"
P4-CC	"
P5-39	"
P6-R	"
P6-M	"
P8-S	"
P9-S	"
P10-E	"
P10-S	"
P10-T	"
1J1-B	"
1J1-F	"
1J1-G	"
1J1-H	"
1J1-J	"
1J1-N	"
1J1-P	"
1J1-R	"
1J1-a	"
1J1-c	"
1J1-f	"
1J1-g	"
1J1-h	"
1J1-j	"
1J2-N	"
1J12-B	"
1J12-h	"

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Table IV (Con't)

From	To	From	To
P19-4K	All others listed & shell	1J12-g	All others listed & shell
P19-4E	"	1J12-E	"
P19-4N	"	1J12-D	"
P19-4P	"	1J12-L	"
P19-4R	"	J3-C	"
P19-4S	"	J3-F	"
P19-4T	"	J3-U	"
P19-4U	"	J3-Y	"
P19-4V	"	J3-a	"
P19-4W	"	J3-k	"
P4-A	"	J3-g	"
P4-B	"	J3-n	"
P4-C	"	J3-s	"
P19-3P	"	1J2-C	"
P19-5B	"	P19-5A	"
P19-5M	"	1J2-c	"
P10-C	"	1J2-b	"
P10-J	"	P19-5J	"
P10-M	"	P19-5K	"
P10-R	"	P4-AA	"
P10-U	"	P5-34	"
P10-b	"	P6-P	"
P19-1T	"	P7-R	"

The following list of unused pins may be connected together and tested as one terminal:

P19-1H, 1J, 3J, 5G, 5H, 5N, 5P, 5R, 5S, 5T, 5U, 5V, 5W, 3K.

P4-L, P, R, d, e, f, g, h, k, m, n, p, q, r, s, t, u, v, x, y, z, EE, FF, GG, HH.

J3-PP.

P21A-B.

P5, 1, 2, 3, 10, 13, 17, 32, 37, 38, 42, 44, 53, 54, 55.

P8, A, F, G, T.

P9, A, F, G, T.

P10, A, B, F, G, H, K, L, P, X, c.

J15, C, J, K, F, T, U.

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1J2, T, Z.

1J12, C, S, A, M, U, a, b, d, W, j, T, P.

1J13, 2, 4, 26, 19, 20, 21, 41.

3.10.2 1 Megohm insulation resistance terminal board and harness assembly drawing 9237021. The insulation resistance between each circuit and all other circuits as indicated in Table V shall not be less than 1 megohm when tested as specified in 4.4.2.2.

Table V

From	To	**From	**To
1J13-13	All others listed* and shell	1J16C	1J16S
P5-30	"	1J17C	1J17S
J3-z	"		
P5-31	"	P19-4C	P19-5E
J3-BB	"	P19-2F	J3-n
J15-D	"	J3-s	P19-5E
P4-T	"	P19-5F	P19-4F
P4-X	"	P10-S	P19-5M
P5-6	"	P10-T	P19-5M
P6-U	"	J3-C	P6-M
P4-G	"		
1J13-3	"		
P5-7	"		
P4-J	"		

*Tables IV and V except **

3.11 Diode tests.--(BCC)

3.11.1 Reverse current leakage. The reverse current leakage of each circuit shall not exceed the values as shown in Table VI when tested as specified in 4.4.3.1.

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Table VI

Diode No. Under Test	+Terminal*	-Terminal*	Applied Reverse Direct Current Voltage ARVDC	Test Code (see 3.5)	Voltage Drop (max) Across Series Resistor at 200 ARVDC (see 4.4)
CR201	J3-M	J3-w	200 ± 10	102001	25
CR202					
CR203	J3-M	J3-c	200 ± 10	102002	25
CR204					
CR207	J3-H	P19-4A	200 ± 10	102003	25
CR208					
CR209	J3-H	P19-4B	200 ± 10	102004	25
CR210					
CR205	1J13-9	1J13-8	200 ± 10	102005	25
CR206	1J13-10	1J13-8	200 ± 10	102006	25
CR218	P4-T	P6-K	200 ± 10	102007	25
CR217	P4-T	P7-K	200 ± 10	102008	25
CR216	P4-K	P6-A	200 ± 10	102009	25
CR215	P4-K	P7-A	200 ± 10	102010	25

3.11.2 Forward voltage drop. The forward voltage drop of each circuit shall not exceed the maximum voltage values as shown in Table VII when tested as specified in 4.4.3.1.

Table VII

Diode No. Under Test	+Terminal*	-Terminal*	Applied Forward Current (AMPS)	Test Code (see 3.5)	Max Forward Voltage Drop Volts (V)
CR201	J3-w	J3-M	5.0 ± 0.5	103001	3.5
CR202					
CR203	J3-c	J3-M	5.0 ± 0.5	103002	3.5
CR204					
CR207	P19-4A	J3-H	5.0 ± 0.5	103003	3.5
CR208					
CR209	P19-4B	J3-H	5.0 ± 0.5	103004	3.5
CR210					
CR205	1J13-8	1J13-9	5.0 ± 0.5	103005	3.5
CR206	1J13-8	1J13-10	5.0 ± 0.5	103006	3.5
CR215	P7-A	P4-K	1.0 ± 0.1	103007	2.0
CR216	P6-A	P4-K	1.0 ± 0.1	103008	2.0
CR217	P7-K	P4-T	1.0 ± 0.1	103009	2.0
CR218	P6-K	P4-T	1.0 ± 0.1	103010	2.0
CR211	1J13-11	1J13-7	5.0 ± 0.5	103011	3.5
CR212	1J13-12	1J13-7	5.0 ± 0.5	103012	3.5

*Polarity of measuring device.

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3.12 Delay circuit test.--(BCC)

3.12.1 Time delay functioning.

3.12.1.1 Non-function. The delay unit shall not produce an output with a control input pulse width of 0.899 seconds +0.000 -0.100 seconds. Testing shall be as specified in 4.4.4.3.1.

3.12.1.2 Function. The delay unit shall function and produce an output when a control input pulse width of 1.00 seconds ± 0.10 is applied. Testing shall be as specified in 4.4.4.3.2.

3.12.1.3 Output. The load current shall be 1.0 ± 0.1 amperes steady state, or 7.0 ± 0.2 for 10.0 ± 1.0 milliseconds. The voltage drop across the solid state switch shall not exceed 4 volts with a current of 1.0 ± 0.1 ampere through the circuit.

3.13 Energy transfer circuit (ETC).--(BCC)

3.13.1 ETM output voltage. The output voltage of channel "A" ETM, and channel "B" ETM shall rise to a minimum of 50 volts in 10 microseconds (ms) maximum. At 20 ms the pulse voltage shall be no less than 25% of the peak voltage. Testing shall be as specified in 4.4.5.

3.14 Circuit resistance after assembly of control unit per dwg 9237001. The resistance of each circuit shall not exceed the maximum value listed in Table VIII or as shown otherwise when tested as specified in 4.4.1. Prior to the performance of this test, the item shall have successfully passed all the test requirements specified in 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, 3.13 and 3.16.

Table VIII

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J3-A	1J13-55	0.3	130001
J3-A	1J13-54	0.3	130002
J3-A	1J13-53	0.3	130003
J3-A	1J13-45	0.3	130004
J3-A	1J13-52	0.3	130005
J3-A	1J12-L	0.3	130006
J3-B	J3-DD	0.3	130007
J3-B	1J13-66	0.3	130008
J3-B	1J12-g	0.3	130009
J3-D	1J2-N	0.3	130010
J3-D	1J2-P	0.3	130011
J3-E	P8-D	0.3	130012
J3-E	P8-B	0.3	130013
J3-E	P8-C	0.3	130014
J3-E	P8-E	0.3	130015
J3-E	P9-B	0.3	130016
J3-E	P9-C	0.3	130017

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Table VIII (Con't)

Terminals From	To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J3-E	P9-D	0.3	130018
J3-E	P9-E	0.3	130019
J3-E	1J13-25	0.3	130020
J3-F	J3-i	0.3	130021
J3-G	1J12-E	0.3	130022
J3-G	1J13-17	0.3	130023
J3-H	1J13-13	0.3	130024
J3-H	J3-p	0.3	130025
J3-J	1J13-49	0.3	130026
J3-J	1J13-44	0.4	130027
J3-J	J15-G	0.4	130028
J3-K	1J12-D	0.3	130029
J3-K	1J13-16	0.3	130030
J3-L	P9-H	21-33	130031
J3-L	P9-J	21-33	130032
J3-L	P9-K	21-33	130033
J3-L	P9-L	21-33	130034
J3-L	P8-H	21-33	130035
J3-L	P8-J	21-33	130036
J3-L	P8-K	21-33	130037
J3-L	P8-L	21-33	130038
J3-M	1J13-3	0.3	130039
J3-M	P4-M	12-16	130040
J3-M	P4-N	12-16	130041
J3-M	P5-9	12-16	130042
J3-M	P5-16	12-16	130043
J3-M	P5-47	12-16	130044
J3-M	P5-48	12-16	130045
J3-M	P6-B	12-16	130046
J3-M	P6-C	12-16	130047
J3-M	P6-E	12-16	130048
J3-M	P6-G	12-16	130049
J3-M	P7-B	12-16	130050
J3-M	P7-C	12-16	130051
J3-M	P7-E	12-16	130052
J3-M	P7-G	12-16	130053
J3-M	1J13-24	12-16	130054
J3-N	J3-Z	0.3	130055
J3-N	1J1-h	0.3	130056
J3-R	P4-D	0.3	130057
J3-S	1J1-R	0.3	130058
J3-T	J3-GG	0.3	130059
J3-T	1J13-67	0.3	130060
J3-T	1J12-h	0.3	130061
J3-U	J3-LL	0.3	130062
J3-V	1J1-j	0.3	130063

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Table VIII (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
J3-V	J3-e	0.3	130064
J3-W	1J1-P	0.3	130065
J3-X	1J1-J	0.3	130066
J3-Y	J3-JJ	0.3	130067
J3-a	J3-NN	0.3	130068
J3-b	1J1-a	0.3	130069
J3-c	P6-H	0.3	130070
J3-c	P6-J	0.3	130071
J3-f	1J1-f	0.3	130072
J3-g	J3-KK	0.3	130073
J3-h	1J1-N	0.3	130074
J3-j	1J1-F	0.3	130075
J3-j	1J1-Z	0.3	130076
J3-k	J3-FF	0.3	130077
J3-m	1J1-g	0.3	130078
J3-q	1J1-c	0.3	130079
J3-r	J3-t	0.3	130080
J3-r	J3-x	0.3	130081
J3-r	J3-y	0.3	130082
J3-R	P4-C	0.3	130083
J3-r	1J12-B	0.3	130084
J3-r	1J12-V	0.3	130085
J3-r	1J13-23	0.3	130086
J3-u	J3-HH	0.3	130087
J3-w	P7-H	0.3	130088
J3-w	P7-J	0.3	130089
J3-MM	1J1-G	0.3	130090
J3-MM	1J1-Y	0.3	130091
P4-A	1J2-H	0.3	130092
P4-A	1J13-30	0.3	130093
P4-B	1J13-29	0.3	130094
P4-E	1J12-R	0.3	130095
P4-E	1J13-22	0.3	130096
P4-F	1J12-e	0.3	130097
P4-G	P4-H	0.3	130098
P4-G	P5-27	0.3	130099
P4-G	P5-33	0.3	130100
P4-G	1J13-57	0.3	130101
P4-J	P4-K	0.3	130102
P4-S	1J13-60	0.3	130103
P4-S	P5-26	0.3	130104
P4-T	P4-U	0.3	130105
P4-V	1J1-S	0.3	130106
P4-V	1J1-V	0.3	130107
P4-V	1J1-C	0.3	130108
P4-V	1J1-d	0.3	130109

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Table VIII (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P4-V	1J2-L	0.3	130110
P4-V	1J2-V	0.3	130111
P4-V	P4-c	0.3	130112
P4-X	P5-11	8-12	130113
P4-X	P5-12	8-12	130114
P4-E	1J1-e	0.3	130115
P4-E	1J1-T	0.3	130116
P4-F	1J1-A	0.3	130117
P4-F	1J1-U	0.3	130118
P4-X	P5-20	12-16	130119
P4-X	P5-21	12-16	130120
P4-X	P5-28	12-16	130121
P4-X	P5-29	12-16	130122
P4-X	1J13-8	12-16	130123
P4-X	P4-Y	0.3	130125
P4-W	1J13-28	0.3	130126
P4-a	1J13-31	0.3	130127
P4-a	P5-24	0.3	130128
P4-AA	P4-BB	0.3	130129
P4-AA	P5-23	12-16	130130
P4-AA	P5-22	12-16	130131
P4-AA	P5-51	12-16	130132
P4-AA	P5-45	12-16	130133
P4-AA	1J13-15	12-16	130134
P4-CC	P4-DD	0.3	130135
P4-CC	1J12-c	0.3	130136
P4-CC	1J12-N	0.3	130137
P4-CC	1J13-18	0.3	130138
P5-5	P8-N	0.3	130139
P5-5	P9-M	0.3	130140
P5-15	1J13-11	0.3	130141
P5-5	P5-7	0.9-2.5	130142
P5-5	P8-P	0.9-2.5	130143
P5-5	P9-U	0.9-2.5	130144
P5-5	1J13-9	0.9-2.5	130145
P5-6	1J13-10	0.3	130146
P5-6	P8-U	0.3	130147
P5-6	P9-P	0.3	130148
P5-6	P9-N	0.9-2.5	130149
P5-6	P8-M	0.9-2.5	130150
P5-6	P5-8	0.9-2.5	130151
P5-14	1J13-12	0.3	130152
P5-18	P5-19	0.3	130153
P5-18	P5-52	1.5-2.5	130154
P5-18	1J13-14	0.3	130155
P5-18	P5-46	1.5-2.5	130156

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Table VIII (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P5-30	1J12-F	0.3	130157
P5-30	1J12-f	0.3	130158
P5-30	1J12-Y	0.3	130159
P5-30	1J12-Z	0.3	130160
P5-31	1J12-G	0.3	130161
P5-31	1J12-H	0.3	130162
P5-31	1J12-J	0.3	130163
P5-31	1J12-X	0.3	130164
P5-35	P5-43	0.3	130165
P5-35	P5-36	0.3	130166
P5-39	1J13-33	0.3	130167
P5-39	P6-S	0.3	130168
P5-49	1J13-27	0.75-1.5	130169
P5-49	P5-50	0.75-1.5	130170
P6-K	P6-T	0.3	130171
P6-F	P6-A	9-20	130173
P6-F	P6-L	9-20	130174
P6-F	P6-D	0.3	130175
P7-K	P7-T	0.3	130177
P6-F	P7-F	0.3	130178
P6-F	P7-D	0.3	130179
P6-F	P7-A	9-20	130180
P6-F	P7-L	9-20	130181
1J13-60	1J13-24	See Note 1	130182
1J13-57	1J13-24	See Note 1	130183
P6-M	P6-N	0.3	130184
P6-M	P7-M	27-35	130185
P6-M	P7-N	27-35	130186
P6-P	P7-P	0.3	130187
P6-P	1J2-K	0.3	130188
P6-P	1J2-U	0.3	130189
P6-U	1J13-50	0.3	130190
P6-U	P7-U	0.3	130191
P6-U	1J2-D	0.3	130192
P6-U	1J2-X	0.3	130193
P6-R	P7-S	0.3	130194
P6-R	1J13-34	0.3	130195
P7-R	P8-R	0.3	130196

Note 1 - Set the Simpson 269 to RX100 scale. Connect the positive lead of the Simpson to the test points listed in the "FROM" column. Connect the negative lead of the Simpson to the test point listed in the "TO" column. The Simpson meter will slowly rise to a high resistance.

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Table VIII (Con't)

Terminals From	Terminals To	Resistance (ohms) (max)	Test Code No. (see 3.5)
P7-R	1J13-36	0.3	130197
P8-S	1J13-37	0.3	130198
P8-S	P9-R	0.3	130199
P9-S	1J13-38	0.3	130200
P9-S	1J13-40	0.3	130201
P9-S	J15-H	0.3	130202
J15-D	J15-E	0.3	130203
J15-D	1J13-48	0.3	130204
J15-G	1J13-44	0.3	130205
J15-L	J15-M	0.3	130206
J15-N	J15-P	0.3	130207
J15-N	1J13-51	0.3	130208
J15-R	1J13-35	0.3	130209
J15-A	1J13-7	0.3	130210
J15-B	1J13-7	0.5	130211
J15-S	1J13-39	0.3	130212
1J1-A	1J1-U	0.3	130213
1J1-B	1J13-47	0.3	130214
1J1-B	1J12-K	0.3	130215
1J1-H	1J2-B	0.3	130216
1J1-H	1J13-42	0.3	130217
1J1-K	1J1-b	0.3	130218
1J1-K	1J13-63	0.3	130219
1J1-L	1J1-M	0.3	130220
1J1-L	1J13-62	0.3	130221
1J1-T	1J1-e	0.3	130222
1J2-A	1J2-J	0.3	130223
1J2-C	1J2-R	0.3	130224
1J2-E	1J13-6	0.3	130225
1J2-E	1J2-Y	0.3	130226
1J2-F	1J2-W	0.3	130227
1J2-G	1J2-M	0.3	130228
1J2-a	1J13-43	0.3	130229
1J2-b	P5-40	0.3	130230
1J2-S	1J13-46	0.3	130231
1J13-61	1J13-64	0.3	130232
1J13-24	P5-35	0.3	130233
1J13-24	P5-36	0.3	130234
1J13-24	P5-43	0.3	130235
P6-D	1J13-11	270-330	130236
P6-D	1J13-12	270-330	130237
P5-25	P5-4	0.3	130238
1J2-c	P5-41	0.3	130239

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3.15 Insulation resistance after assembly of control unit per dwg 9237001. The insulation resistance between each circuit and all other circuits, as indicated in Table IX, shall not be less than 100 megohms when tested as specified in 4.4.2. Prior to the performance of this test, the item shall have successfully passed all the test requirements specified in 3.7, 3.8, 3.9, 3.10, 3.11, 3.12, 3.13 and 3.16.

Table IX

From	To	From	To
J3-A	All other listed circuits	J3-b	All other listed circuits
J3-B	"	J3-d	"
J3-C	"	J3-f	"
J3-D	"	J3-g	"
J3-E	"	J3-h	"
J3-F	"	J3-j	"
J3-G	"	J3-k	"
J3-J	"	J3-n	"
J3-K	"	J3-q	"
J3-N	"	J3-r	"
J3-P	"	J3-s	"
J3-R	"	J3-v	"
J3-S	"	J3-AA	"
J3-T	"	J3-MM	"
J3-U	"	P4-CC	"
J3-W	"	P4-E	"
J3-Y	"	P4-B	"
J3-a	"	1J1-B	"
P4-F	"	1J1-H	"
P4-Z	"	1J1-K	"
P4-a	"	1J1-L	"
P4-b	"	1J2-A	"
P4-W	"	1J2-C	"
P6-M	"	1J2-F	"
P6-U	"	1J2-G	"
P6-P	"	1J2-b	"
P7-S	"	1J13-5	"
P8-S	"	1J2-c	"
P8-R	"	P5-25	"
P9-S	"	1J13-56	"
P4-A	"	1J13-61	"

The following list of unused pins may be connected together and tested as one terminal:

J3-PP
P4-L, P, R, d, e, f, g, h, k, m, n, p, q, r, s, t, u, v, w, x, y, z, EE, FF, GG, HH.

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P5-1, 2, 3, 10, 13, 17, 32, 34, 37, 38, 42, 44, 53, 54, 55.
P8-A, F, G, T.
P9-A, F, G, T.
J15-C, F, J, K, T, U.
1J2-T, Z.
1J13-2, 4, 26, 41, 19, 20, 21

All connector shells shall be connected together as one terminal.

3.16 Relay functioning.--(BCC)

3.16.1 Pull-in voltage. The pull-in voltage shall be a maximum of 18.0 vdc. Testing shall be as specified in 4.4.6.1.

3.17 Temperature shock. The Control Unit shall withstand a temperature shock with temperatures ranging from -65 to 160°F without evidence of cracking, splitting, separation of parts, or other damage. Testing shall be as specified in 4.4.7. At the completion of the final cycle, the Control Unit shall comply with the requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.18 Vibration.

3.18.1 Transportation. The Control Unit shall withstand the vibration conditions specified in Table X along each of its three mutually perpendicular axes (see Figure 2). Frequency shall be cycled at a logarithmic sweep rate between 5-500-5 Hertz (Hz) in 15 minutes. Testing shall be as specified in 4.4.8. After this test, the Control Unit shall comply with the requirements of 3.14 and 3.15 when tested in accordance with the applicable paragraphs of 4.4 respectively.

Table X

Frequency (Hz)	Gravity Units (g's) or Double Amplitude (da) -65 + 5°F	160 + 5°F	Sweep rate
5-17-5	0.2" da ± 0.02	0.2" da ± 0.2	3 Frequency sweeps from 5-500-5 (15 Minutes each sweep) for a total of 45 minutes per axis.
17-60-17	3.0 g's ± 0.03	3.0 g's ± 0.03	
60-80-60	10.0 g's ± 1.0g	10.0 g's ± 1.0g	
80-100-80	20.0 g's ± 2.0g	20.0 g's ± 2.0g	
100-500-100	5.0 g's ± 0.5g	5.0 g's ± 0.5g	

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3.18.2 Operation. The Control Unit shall be vibrated in each of three mutually perpendicular axes (see Figure 2), after being stabilized at -65 and 160°F as applicable. The inputs shall be applied in the following sequence and conducted as specified in 4.4.

0.3 g^2/Hz power spectral density (19.0 root mean square (rms) g's min) at frequencies ranging from 100 to 1000 Hz, attenuated 12 decibel (db)/octave so that the total frequency envelope is 50 to 2000 Hz for 10 seconds. Tolerance level: ± 3 db between frequencies of 100 and 1000 Hz and ± 6.0 db between frequencies of 1000-2000 Hz.

0.06 g^2/Hz power spectral density (8.5 rms g's min) at frequencies ranging from 100 to 1000 Hz, attenuated 12 db/octave so that the frequency envelope is 50 to 2000 Hz for one minute. Tolerance level: ± 3 db between frequencies of 100-2000 Hz.

After completion of operational vibration test, the Control Unit shall be returned to ambient temperature and shall then meet the initial requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.19 Mechanical shock. The Control Unit shall withstand a shock of 40 ± 4.0 gravity units (g's) along its longitudinal axis in the + x direction (see Figure 2) and 25 ± 2.5 g's both directions along each of its transverse and vertical axes (see Figure 2). The shocks shall be half-sine wave shape and shall be 11 ± 1.0 milliseconds (ms) duration along the longitudinal axis and 8 ± 0.8 ms duration along the transverse and vertical axes. Testing shall be as specified in 4.4.12. Upon completion of the test the Control Unit shall meet the requirements of 3.14 and 3.15 when tested in accordance with the applicable test paragraphs of 4.4 respectively.

3.20 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.21 Workmanship.

3.21.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.21.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.21.3 Burrs. Parts shall be free from burrs.

3.21.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.21.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.21.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 4 complete assemblies (see dwg 9237001) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table XI. Tests shall be conducted in order specified in Table XI.

Table XI

First article inspection and tests
Control Unit No.

Inspection or Tests	Requirement	1	2	3	4
1. Classification of Defects/ Characteristics	4.3.2.1 (except 4.3.2.1.4)	X	X	X	X
2. Circuit Resistance Diodes	3.11	X	X	X	X
3. Delay Unit Test	3.12	X	X	X	X
4. Relay Test	3.16	X	X	X	X
5. Energy Transfer Test	3.13	X	X	X	X
6. Classification of Defects/ Characteristics	4.3.2.1.4	X	X	X	X
7. Circuit Resistance	3.14	X	X	X	X
8. Insulation Resistance	3.15	X	X	X	X

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Table XI (Con't)

Inspection or Tests	Requirement	1	2	3	4
9. Temperature Shock	3.17	X	X	X	X
10. Vibration @ (160°F)	3.18	X	X	X	X
11. Vibration @ (-65°F)	3.18	X	X	X	X
12. Mechanical Shock	3.19	X	X	X	X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation. The formation of an inspection lot and related functions shall be in accordance with Section 5, MIL-STD-105. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor defects/characteristics table (see Table XII and Table XIII), and inspected for all defects/characteristics in the applicable classification. Any defects/characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of defects/characteristics in each classification need not be followed as long as all defects/characteristics inspections are performed.

Table XII

Major Defects/Characteristics

Lot Size	Sample Size
1-155	100%
156-200	155
201-250	175
251-300	190
301-350	200
351-400	210
401-450	220
451-500	225
501-550	230

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Table XII (Con't)

Lot Size	Sample Size
551-600	235
601-650	240
651-700	243
701-750	247
751-800	250
801-900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Table XIII

Minor Defects/Characteristics

Lot Size	Sample Size
1-25	100%
26-30	25
31-35	26
36-40	27
41-45	28
46-50	29
51-55	30
56-60	31
61-65	32
66-70	33
71-75	34
76-85	35
86-95	36
96-110	37
111-130	38
131-156	39
157-190	40
191-240	41
241-340	42
341-520	43
521-1120	44
1121 and over	45

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4.3.2.1 Classification of defects/characteristics (see 3.2). Critical, major and minor defects/characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Cover (see dwg 9236990) before assembly.

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mislocation of holes (36)	Gage
	102. Incorrect intermediate size diameter clearance holes in flange (10)	Gage
	103. Incorrect largest diameter clearance holes in flange (2)	Gage
	104. Incorrect inside diameter	Gage
	105. Incorrect outside flange diameter	Gage
	106. Mislocation of cutout in mounting flange	Gage
	107. Evidence of poor workmanship (see 3.21)	Visual
	108. Incorrect external radius (max)	Gage
	109. Incorrect width of cutout in mounting flange	Gage
	110. Damage to protective finish exposing base metal	Visual
	111. Incorrect thickness of flange	Gage
	112. Incorrect flatness of flange	Gage
	113. Incorrect size of large cutout	Gage
	114. Part No., incorrect, illegible or missing	Visual
	115. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:	None defined.	

4.3.2.1.2 Plate, connector assembly (see dwg 9237019).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mislocation of clinch nut mounting holes (11) (see dwg 9237017)	Gage

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102. Improper assembly of clinch nuts (11).....Visual/Manual
103. Mislocation of connector mounting holes (6) (see dwg 9237017).....Gage
104. Incorrect size of plate (see dwg 9237017).....Gage
105. Evidence of poor workmanship (see 3.21).....Visual
106. Incorrect thickness (see dwg 9237017).....Gage
107. Damage to protective finish exposing base metal (see dwg 9237017).....Visual
108. Marking illegible, missing or incorrect.....Visual
109. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.3 Mounting plate assembly (see dwg 9236980) inserts installed.

Categories	Defects/Characteristics	Method of Inspection

Critical: None defined.

Major:
101. Improper assembly of inserts (55 places).....Visual/Manual
102. Evidence of poor workmanship (see 3.21).....Visual
103. Objective evidence for casting missing or incomplete.....Visual

For the characteristics listed below, see Plate mounting (dwg 9236970).

104. Mislocation of all tapped, clearance and alignment holes (86).....Gage
105. Incorrect chamfer on outside diameter.....Gage
106. Incorrect width of cutout on mounting diameter.....Gage
107. Incorrect diameter of c'sink (Hole No. 1).....Gage
108. Incorrect diameter of clearance hole on surface "C" (No. 7).....Gage

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109. Incorrect diameter of counterbore (No. 7).....Gage
110. Incorrect depth of counterbore (No. 7).....Gage
111. Incorrect diameter of hole for connector alignment pins (12) (Nos. 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86).....Gage
112. Incorrect inside diameter of "O" Ring groove (4) Nos. 32, 33, 35, 36).....Gage
113. Incorrect width of "O" Ring groove (4) (Nos. 32, 33, 35, 36).....Gage
114. Incorrect depth of "O" Ring groove (4) (Nos. 32, 33, 35, 36).....Gage
115. Incorrect concentricity of "O" Ring groove (4) (Nos. 32, 33, 35, 36).....Gage
116. Incorrect inside diameter of "O" Ring groove (No. 38).....Gage
117. Incorrect width of "O" Ring groove (No. 38).....Gage
118. Incorrect depth of "O" Ring groove (No. 38).....Gage
119. Incorrect concentricity of "O" Ring groove (No. 38).....Gage
120. Incorrect thickness of annular zone between outside diameter and 13.40 diameter.....Gage
121. Incorrect size of larger "D" shaped hole (No. 39).....Gage
122. Incorrect distance from connector mounting surface to datum surface "B" (6) (Nos. 32, 33, 34, 35, 36, 38).....Gage
123. Incorrect connector mounting flange thickness (8) (Nos. 32, 33, 34, 35, 36, 37, 38, 39).....Gage
124. Incorrect flatness of surface "B".....Gage
125. Incorrect outside diameter.....Gage
126. Incorrect flatness of extended boss surfaces.....Gage
127. Incorrect parallelism of surface "C".....Gage
128. Incorrect parallelism of mounting surface (boss side).....Gage

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129. Mislocation of outside diameter cutout.....Gage
130. Incorrect diameter of mounting clearance holes (6) (Nos. 63, 64, 65, 66, 67, 68).....Gage
131. Incorrect diameter of alignment pin holes (2) on mounting surface (Nos. 73 and 74).....Gage
132. Incorrect chamfer of alignment pin holes (2) (Nos. 73 and 74).....Gage
133. Improper surface finish.....Visual
134. Damage to protective finish exposing base metal.....Visual
135. Incorrect outside groove diameter.....Gage
136. Incorrect concentricity of outside groove diameter.....Gage
137. Incorrect width of groove.....Gage
138. Incorrect depth of groove.....Gage
139. Incorrect diameter (Hole No. 2).....Gage
140. All bosses and standoffs covered with heat shrinkable tubing (see 9236980).....Visual
141. All tubing neatly trimmed at ends of bosses and standoffs and no cracks, chips, or tears in tubing (see dwg 9236980).....Visual
142. Improper assembly of spring pin in Hole No. 2 (see 9236980).....Gage
143. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.4 Control unit (see dwg 9237001).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cover gasket improperly bonded to mounting plate assy (cover removed)	Visual/Manual
102.	Incorrect torque on connectors 14 places	Gage
103.	Connector plate gasket missing	Visual
104.	Incorrect torque on cover mounting screws (13 places)	Gage

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105.	Incorrect torque on stand-offs (10 places) (cover removed)	Gage
106.	Incorrect torque for connector receptacle (cover removed)	Gage
107.	Incorrect torque for Gyro switch - 4 places (cover removed)	Gage
108.	Improper bonding of sealing disc	Visual/Manual
109.	Any component hardware or gasket, missing or loose or improperly assembled (cover removed)	Visual/Manual
110.	Evidence of poor workmanship (see 3.21)	Visual
111.	Serialization improper or missing (see 3.4)	Visual
112.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.5 Printed wiring board assembly (see dwg 9237022).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Improper or incorrect installation of squib switches (18)	Visual
102.	Improper or incorrect installation of resistors (18)	Visual
103.	Improper or incorrect installation of insulators (18)	Visual
104.	Cable strapped down securely	Visual/Manual
105.	Cable clear of mounting holes (2 places)	Visual
106.	Incorrect torque for strap mounting device and connector mounting bracket (9)	Gage
107.	Incorrect or improper conformal coating	Visual
108.	Improper soldering (see 3.20)	Visual
109.	Marking improper, incorrect or illegible	Visual
110.	Evidence of poor workmanship (see 3.21)	Visual

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*111.	Circuit resistance incorrect (see 3.7)	Gage
*112.	Insulation resistance incorrect (see 3.8)	Gage
113.	Objective evidence relating to material missing or improper (see 3.1)	Visual

Minor: None defined.

4.3.2.1.6 Terminal board and harness assembly (see dwg 9237021).

Categories	Defects/Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Any component mislocated, incorrect or improperly installed (see dwg 9237028)	Visual
102.	Incorrect torque for installation of diodes (12) (see dwg 9237028)	Gage
103.	Incorrect torque for installation of relays (8) (see dwg 9237028)	Gage
104.	Incorrect torque for installation of delay unit (3 places)	Gage
105.	Incorrect torque for cabling mounting device (9) (see dwg 9237028)	Gage
106.	Improper soldering (see 3.20)	Visual
107.	Jumpers from top to bottom side of board not passing through grommets (see dwg 9237028)	Visual
108.	Any connector incorrect (16)	Visual
109.	Cable strapped down securely	Visual/Manual
110.	Incorrect or improper conformal coating	Visual
111.	Any component missing, loose or damaged	Visual/Manual
112.	Marking improper, illegible or incorrect	Visual
113.	Evidence of poor workmanship (see 3.21)	Visual
*114.	Circuit resistance incorrect (see 3.9)	Gage

*100% inspection.

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*115. Insulation resistance incorrect (see 3.10).....Gage
116. Incorrect routing of wires.....Visual
117. Incorrect torque for installation of ETM 94 (see dwg 9237028).....Gage
118. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor: None defined.

4.3.2.1.7 Preparation for delivery (see 5).

Categories	Defects/characteristics	Method of Inspection

Critical: None defined.

Major:
101. Preservation, packing, and packaging not in accordance with MIL-STD-1169.....Visual
102. Marking not in accordance with MIL-STD-1169.....Visual

Minor:
201. Evidence of poor workmanship (see 3.21).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Circuit resistance and insulation resistance. Each Control Unit in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.14 and 3.15, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major defects/characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL).

*100% inspection.

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ELs for assemblies are listed on the Equipment Tabulation, ET-9257001. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor defects/characteristics and tests. Acceptance inspection equipment for minor defects/characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance.

4.4.1.1 Circuit resistance. The circuit resistance of each board (Printed & Terminal) and the assembled Control Unit shall be measured by utilizing a suitable circuit resistance tester. The measured resistance shall be specified in 3.7, 3.9 and 3.14. Current shall be limited to 10 milliamperes (ma) ± 0.10 ma, unless otherwise specified.

4.4.1.2 Circuit resistance. The circuit resistance shall be measured using a Simpson VOM Model 269. The measured resistance with the scale setting shall be as specified in 3.14.

4.4.2 Insulation resistance.

4.4.2.1 100 Megohm insulation resistance. An insulation resistance tester, capable of supplying 500 volts direct current (vdc) ± 50 (vdc), and with which the required resistance in megohms can be verified, shall be used to measure the insulation resistance. The current shall be limited to 10 ma (max). The measured resistance shall be as specified in 3.8, 3.10.1 and 3.15. The test voltage shall be applied for 5 seconds, minimum. The test time shall not exceed 60 seconds.

4.4.2.2 1 Megohm insulation resistance. An insulation resistance tester, capable of supplying 20 volts maximum direct current (vdc), and with which the required resistance in megohms can be verified, shall be used to measure the insulation resistance. The current shall be limited to 10 ma (max). The measured resistance shall be as specified in 3.10.2. The test voltage shall be applied for 5 seconds minimum. The test shall not exceed 60 seconds.

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4.4.3 Diode tests.

4.4.3.1 Diode characteristics (reverse current leakage and forward voltage drop). Diodes shall be tested for forward voltage drop when the current specified in 3.11.2 is applied in the forward direction. Reverse current shall be determined by the voltage drop across a 1.0 ± 0.1 megohm resistor when the voltage is applied in the reverse direction as specified in 3.11.1. The duration of the applied signal in either the forward or the reverse direction shall not be more than 15 seconds.

4.4.4 Delay unit test.

4.4.4.1 Power input. The power input to the delay unit shall be as follows: Ebb +30 vdc ± 5 vdc; scr +30 vdc +5 vdc -4 vdc. Power will be applied simultaneously to pins 9, 8 and 1 and shall be a square type input voltage with a rise time of 3 ± 2 microseconds.

4.4.4.2 Control input voltage. The control input voltage to the delay unit shall be from +12 vdc to +35 vdc and shall be square type input voltage with a rise time of 5 ± 3 microseconds.

4.4.4.3 Functioning.

4.4.4.3.1 Nonfunction. The delay unit shall be electrically connected as shown in Figure 1 and with control input pulse as specified in 3.12.2. The unit shall not produce any output. Power input shall be as specified in 4.4.4.1. Control input voltage as specified in 4.4.4.2.

4.4.4.3.2 Function. The delay unit shall be functioned in accordance with 3.12.1.2 when electrically connected as shown in Figure 1. Power shall be applied as specified in 4.4.4.1.

4.4.5 ETM output voltage. An external load of 1.0 ± 0.1 ohm, 1 watt non-inductive resistor connected in series with a 0.5 ± 0.02, 1 watt non-inductive resistor shall be connected across the output terminal of each channel ("A & B") as shown in Figure 3. Momentarily depress SW-1 and record output voltage across external 1.0 ohm load for channel "A". Record for channel "B" by momentarily depressing SW-2. This method, in conjunction with Figure 3, is a suggested method. Any other method which is equivalent will be acceptable, contingent on Government approval.

4.4.6 Relay functioning.

4.4.6.1 Pull-in voltage. A suitable dc voltage measuring device shall be connected to the coil terminals and shall be used to measure the pull-in voltage which will consistently operate the relay. Three consecutive pull-ins shall be considered consistent performance. See Figure 4 for coil and contact terminals.

4.4.7 Temperature shock. The Control Unit shall be tested in accordance with MIL-STD-810, Method 503. At the completion of tests, the Control Unit shall meet the requirements of 3.17.

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4.4.8 Vibration. The Control Unit shall be mounted securely to a suitable fixture and vibrated at the levels and conditions specified in 3.18.1. After undergoing the operational vibration, the Control Unit shall meet the post-test requirements of 3.18.

4.4.9 Mechanical shock. The Control Unit assembly shall be mounted to a suitable fixture and shocked at the levels and conditions specified in 3.19. After undergoing the mechanical shock, the Control Unit shall meet the post-test requirements of 3.19.

4.4.10 Operational vibration. The Control Unit shall be mounted to a suitable fixture and vibrated at the levels and conditions specified in 3.18.2. After undergoing vibration, the Control Unit shall meet the post-test requirements of 3.18.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The Control Unit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

5.1.2 Level C. The Control Unit shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(PA).

5.2 Packing.

5.2.1 Levels A and C. The Control Unit shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(PA).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(PA).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 3.5). Data specified in 3.5, 3.7, 3.9, 3.11 and 3.14 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest and lowest) continually observed over 5 minutes min do not exceed the nominal temperature plus tolerance specified.

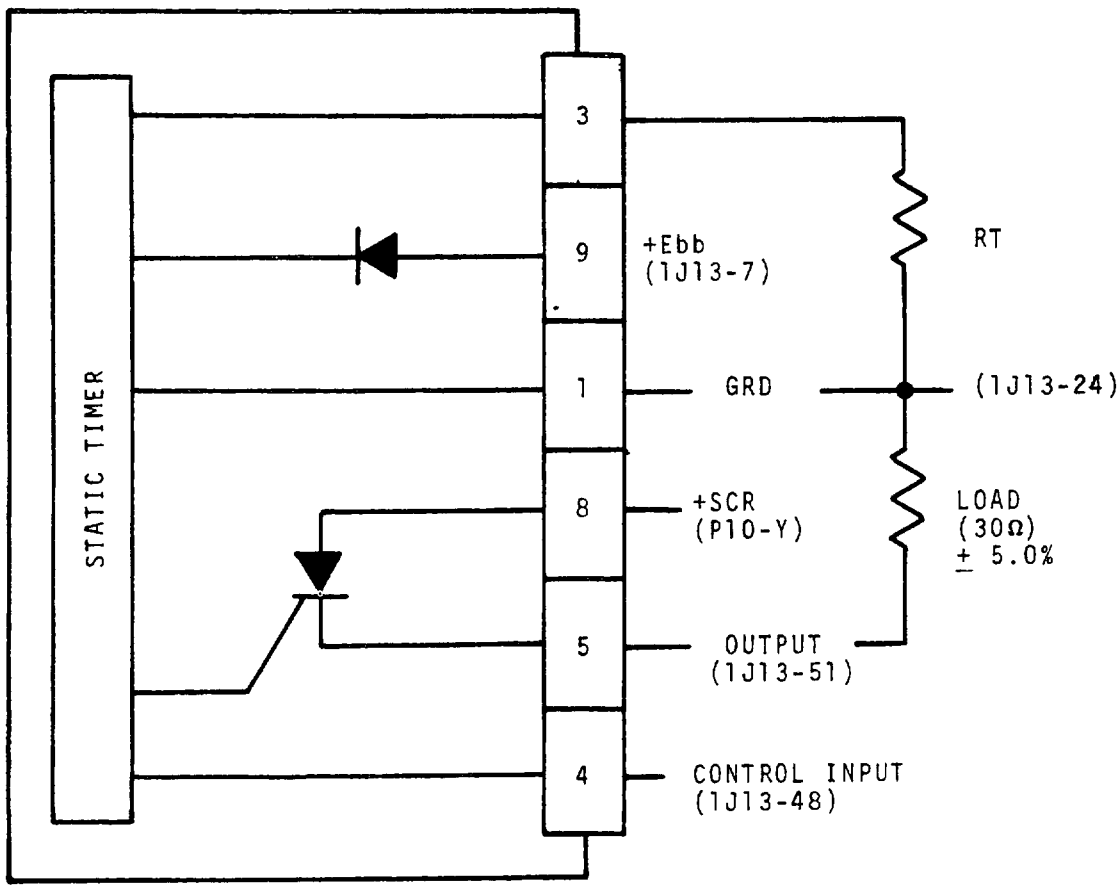
6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1205B Amendment 7 dated 8 July 1976.

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R_T = 47KΩ ± 5%

Figure 1. Delay Unit

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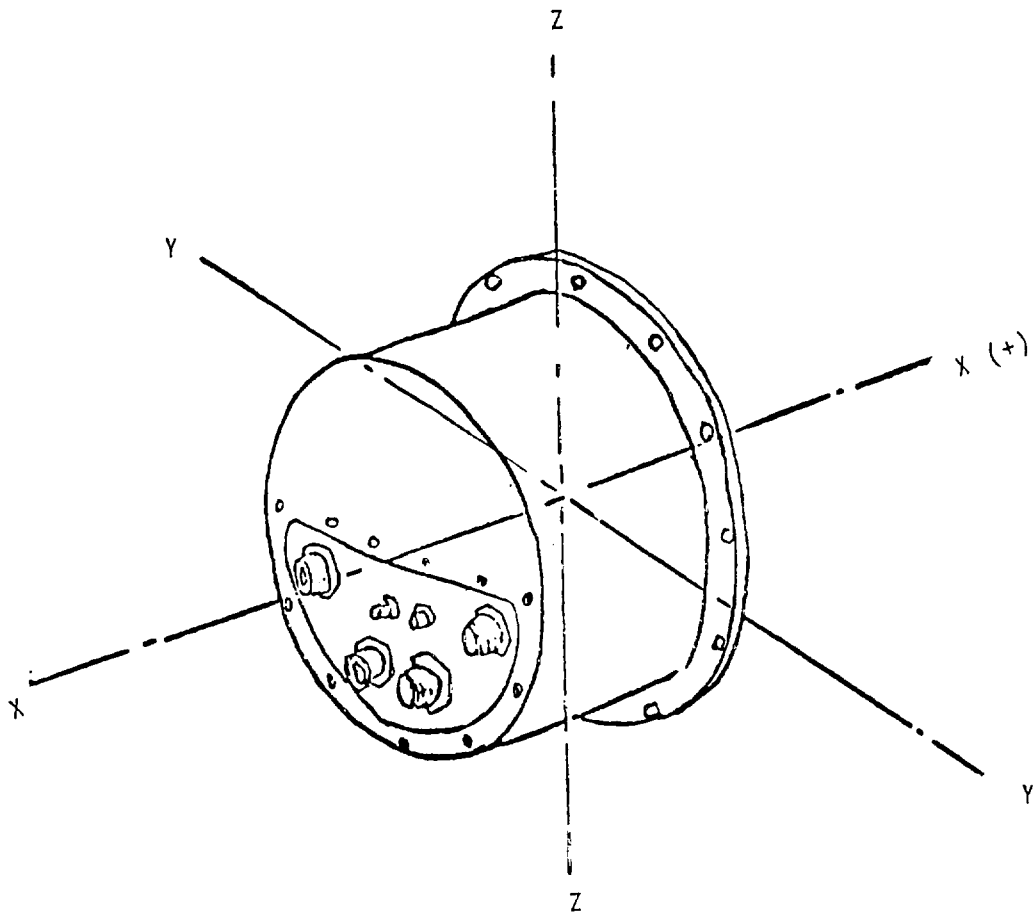


Figure 2. Control Unit

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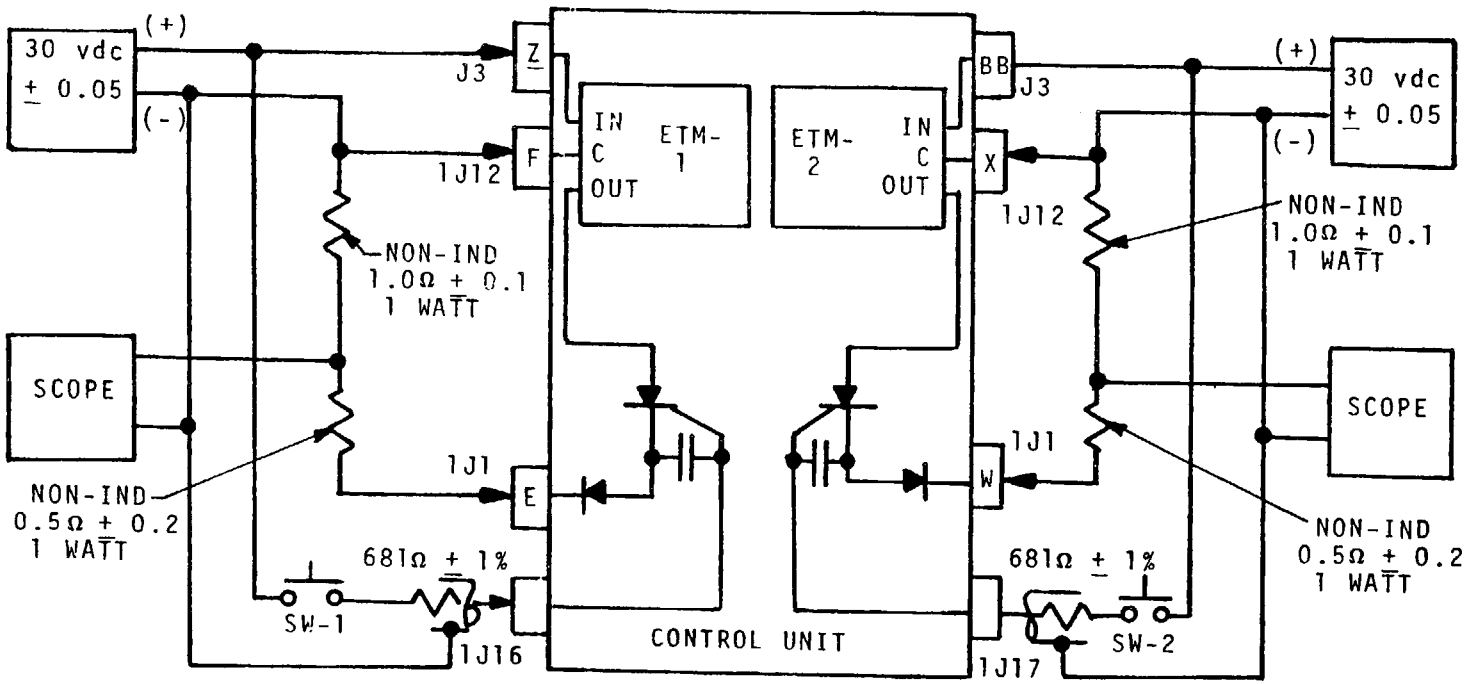
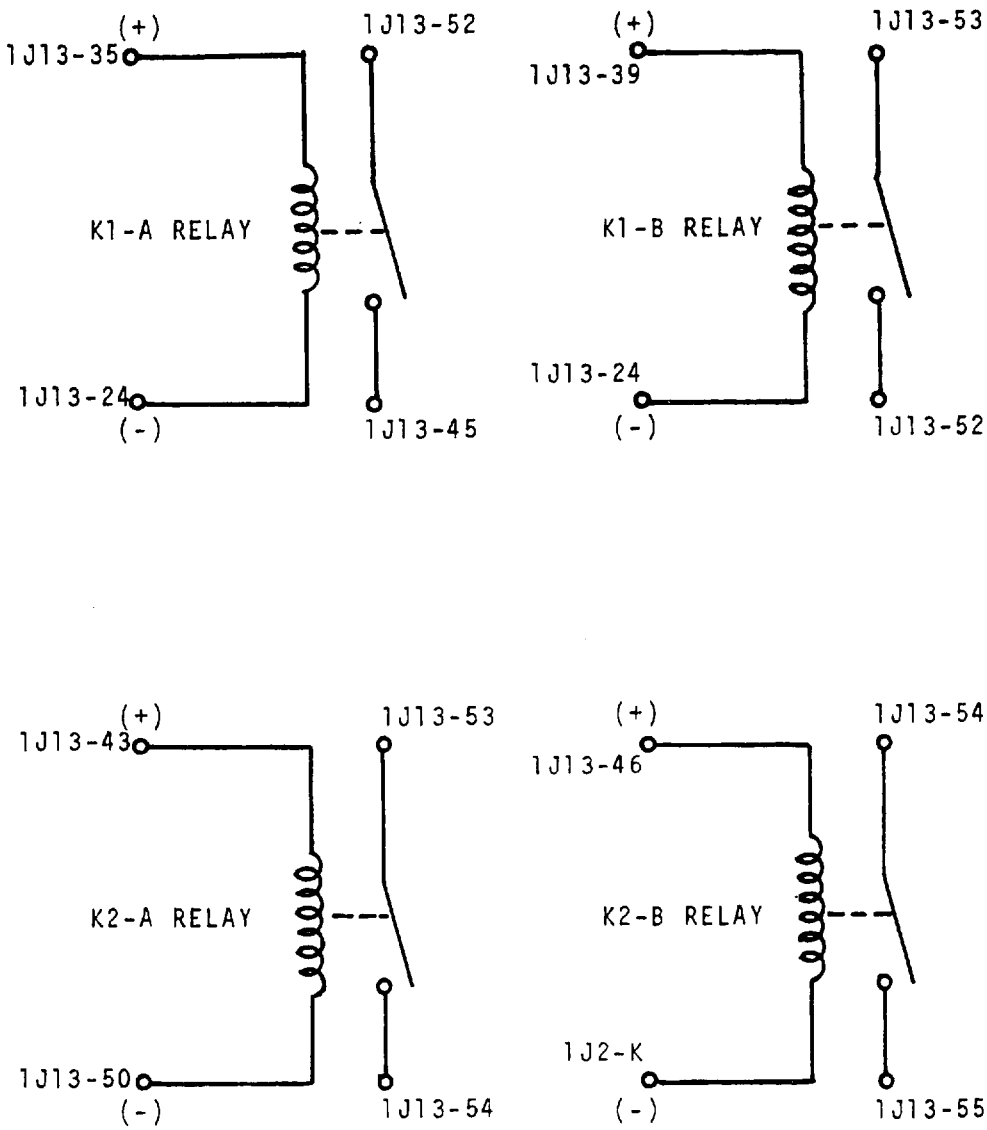


Figure 3. Energy Transfer Module Test Setup

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NOTE: RELAYS SHOWN ARE IN THE ENERGIZED POSITION.

Figure 4. K1, K2 Relays

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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